

Fast First-Order Effective-Rank-Constrained Update

Formula-preserving reference

Core update

The formulas below define a first-order update for a weight matrix while maintaining a scheduled normalized effective-rank floor. They introduce the effective-rank quantity, its local boundary normal and threshold, and then select the spectral-norm-bounded update direction that maximizes alignment with the loss gradient.

$$\begin{aligned}
 W_t, G_t, \Phi_t &\in \mathbb{R}^{m \times r}, \quad G_t = \nabla_{W_t} \mathcal{L}(W_t), \quad m \geq r, \\
 Y_t &= W_t - \eta_t \Phi_t, \quad \mathcal{E}(Y) = \frac{\|Y\|_F^4}{r \|Y^\top Y\|_F^2}, \quad \mathcal{E}(Y_t) \geq \rho_t, \\
 \rho_t &= \rho_0 + (\rho_T - \rho_0) \frac{t}{T}, \quad \chi_t = \sqrt{\rho_t r}, \\
 f_t(W_t) &= \chi_t \|W_t^\top W_t\|_F - \|W_t\|_F^2, \\
 A_t &= 2 \left[\chi_t \frac{W_t(W_t^\top W_t)}{\|W_t^\top W_t\|_F} - W_t \right], \\
 f_t(W_t - \eta_t \Phi) &= f_t(W_t) - \eta_t \langle A_t, \Phi \rangle_F + O(\eta_t^2), \\
 b_t &= \frac{f_t(W_t)}{\eta_t}, \quad \mathcal{E}(W_t - \eta_t \Phi) \geq \rho_t \approx \langle A_t, \Phi \rangle_F \geq b_t, \\
 \Phi_t^* &\in \arg \max_{\|\Phi\|_2 \leq 1, \langle A_t, \Phi \rangle_F \geq b_t} \langle G_t, \Phi \rangle_F, \\
 \mathcal{L}_t(\Phi, \mu) &= \langle G_t, \Phi \rangle_F + \mu (\langle A_t, \Phi \rangle_F - b_t), \quad \mu \geq 0, \\
 B_t(\mu) &= G_t + \mu A_t, \\
 \Phi_t(\mu) &= \text{msign}(G_t + \mu A_t), \\
 d_t(\mu) &= \|G_t + \mu A_t\|_* - \mu b_t, \quad \mu_t^* \in \arg \min_{\mu \geq 0} d_t(\mu), \\
 d'_t(\mu) &= \langle A_t, \text{msign}(G_t + \mu A_t) \rangle_F - b_t, \\
 \mu_t^* &= 0 \quad \text{if} \quad \langle A_t, \text{msign}(G_t) \rangle_F \geq b_t, \\
 \mu_t^* &> 0 \implies \langle A_t, \text{msign}(G_t + \mu_t^* A_t) \rangle_F = b_t, \\
 \boxed{\Phi_t^* &= \text{msign}(G_t + \mu_t^* A_t), \quad W_{t+1} = W_t - \eta_t \Phi_t^*}.
 \end{aligned} \tag{1}$$

The scalar dual variable μ_t^* is the only quantity searched over in this characterization. For a trial value of μ , the displayed matrix-sign expression directly determines the candidate update direction.

Safe finite-step version

The following formulas replace the first-order threshold with a finite-step-safe threshold. The added term supplies the stated safety margin, while the same matrix-sign update form is retained.

$$\begin{aligned}
b_t^{\text{safe}} &= b_t + (3\sqrt{\rho_t r} - 1) \eta_t r, \\
\langle A_t, \Phi \rangle_F \geq b_t^{\text{safe}} &\implies \mathcal{E}(W_t - \eta_t \Phi) \geq \rho_t, \\
d_t^{\text{safe}}(\mu) &= \|G_t + \mu A_t\|_* - \mu b_t^{\text{safe}}, \\
\mu_t^* &\in \arg \min_{\mu \geq 0} d_t^{\text{safe}}(\mu), \\
\boxed{\Phi_t^* &= \text{msign}(G_t + \mu_t^* A_t), \quad W_{t+1} = W_t - \eta_t \Phi_t^*}.
\end{aligned} \tag{2}$$

Variables and computation

Here W_t is the current weight matrix, G_t its loss gradient, Φ_t the update direction, and η_t the learning rate. The schedule ρ_t is the normalized effective-rank floor; A_t and b_t express its first-order boundary condition; b_t^{safe} is the finite-step-safe threshold. The norms $\|\cdot\|_2$, $\|\cdot\|_F$, and $\|\cdot\|_*$ are respectively the spectral, Frobenius, and nuclear norms.

The computational structure is

$$\boxed{W_t, G_t \longrightarrow A_t, b_t \longrightarrow \mu_t^* \longrightarrow \Phi_t^* \longrightarrow W_{t+1}.} \tag{3}$$

There is no inner nonlinear iteration over Φ : for every scalar trial μ ,

$$\boxed{\Phi(\mu) = \text{msign}(G_t + \mu A_t).} \tag{4}$$